UHF Antenna Choices

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Abstract

The implementation of DTV is forcing every television broadcaster to purchase a new antenna. Choosing the best antenna to meet all of the station's needs is not a simple task. With tower space at a premium, tower leasing rates soaring, and FCC deadlines approaching quickly, the broadcaster faces new and non-traditional issues - for example:

- Do I co-locate with others?
- Do I go on the air with a temporary low power system now and worry about my final higher power system later?
- Can I or should I broadcast my current NTSC signal and my new DTV signal on the same antenna?

These questions have no historic precedent. To answer them, the broadcaster needs to know about the types of antennas available and the advantages and concerns associated with each one.

Three types of UHF antennas are on the market today. They are the slot, the panel, and the superturnstile. This paper presents an overview of each of these. Their advantages and disadvantages are discussed, and a decision tree is developed to help choose the best antenna.

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UHF Slot Antennas

Domestically, the best-known UHF broadcast antenna type is the slot antenna. Most manufacturers offer configurations that can be used to broadcast a wide variety of azimuth and elevation patterns at either low or high power. With their simple feed systems, low windloads, and ability to be side- or top-mounted, slots are very versatile and popular antennas. Their disadvantages include narrow bandwidth and pattern distortion when sidemounted.

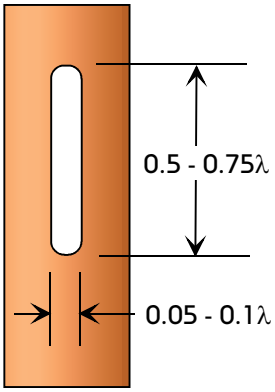


Figure 1. Typical Coax Slot Dimensions

Types

There are two types of slot antennas; standing wave and traveling wave. A standing wave slot antenna has its radiating elements spaced by one wavelength. Thus, at the center frequency of the channel, the slots all radiate in the same phase and amplitude. Standing wave antennas are the most widely used slot antennas today.

A traveling wave antenna, on the other hand, has its slot spacing unequal to the wavelength. Thus the energy radiating from the slots is progressively attenuated, leaving a small amount of energy at the top of the antenna. This excess is either dissipated in a load or radiated through a set of special slots in the top.

In either case, the slots are some fraction of a wavelength long (usually between 0.5 and 0.75 lambda - see figure 1). The RF energy flowing in the antenna is coupled to the slots by the use of probes. These probes, known as couplers, protrude into the space between the inner and outer conductor. There are many different styles and shapes of couplers, but all of them couple energy to the slot.

Operating Principle

Figure 2 shows a cross-section of a coaxial slot antenna. The current travels up the inner conductor, generating an electric field (E-field) and a magnetic field (H-field) between the conductors as in any coax line. The coupler, intercepting these fields, develops an induced axial current, which sets up a circumferential current around the outside of the outer conductor, which in turn creates a voltage potential (an E-field) across the slot. The interaction of this E-field with the outer conductor creates a broadcast signal. The diameter of the outer and the location of the slots dictate the resulting radiation pattern.

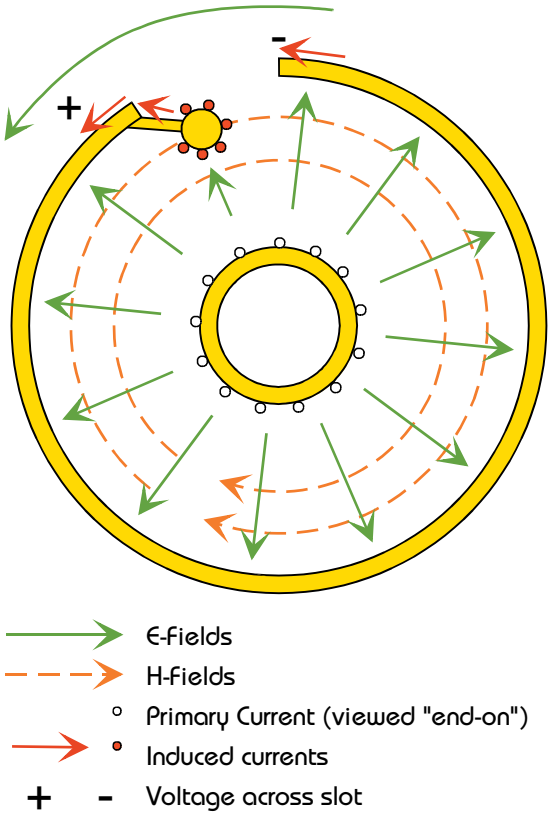


Figure 2. Coaxial Slot Antenna Cross-Section

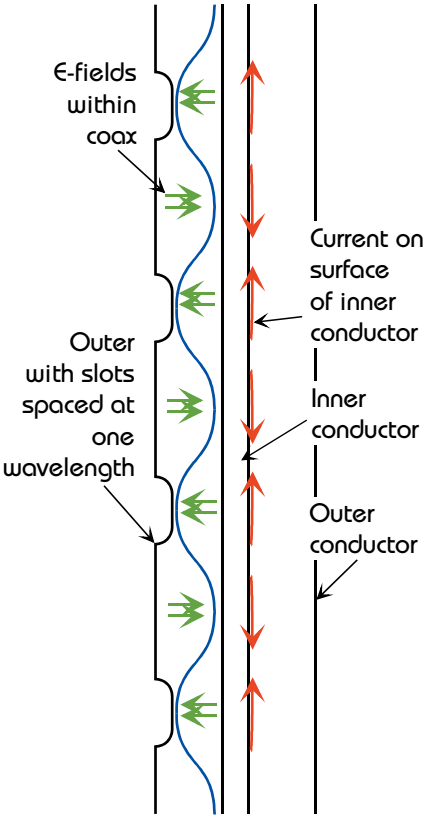
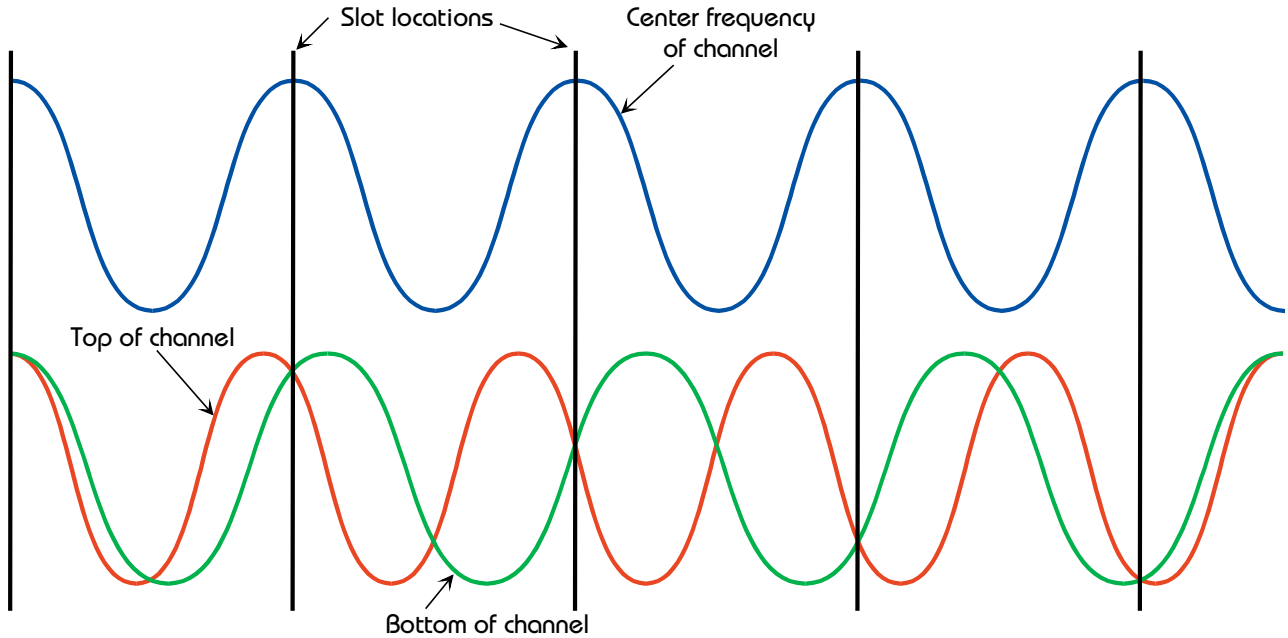


Figure 3. Standing Wave in a Slot Antenna

Figure 3 shows the standing wave inside the antenna at the center frequency of the channel. The slots radiate in phase and at the same amplitude when spaced at 1 wavelength.

Because the channel is 6 MHz wide, and the slot spacing is based on the center frequency, the frequencies at the top and bottom of the channel will radiate at a different phase and amplitude at each slot (see figure 4). In a small (eg: 4-bay) antenna, this does not create a problem, but in a large (eg: 32-bay) antenna, it can result in unwanted beam steering (see figure 5).



• The upper curve represents the center frequency of the channel, to which the slots are spaced. The phase and amplitude of the center frequency remain constant at each slot over the full length of the antenna.

• The phase and amplitude of the lower curves, representing the top of channel (shortest wavelength) and the bottom of channel (longest λ) vary from slot to slot over the full length of the antenna, resulting in beam steering.

Above. Figure 4. Phase Differential Across the Channel (exaggerated)

Shaping the Elevation Pattern

Elevation pattern shaping can be achieved in several ways. One way is to use a traveling wave antenna, with slot spacing calculated to give the desired pattern; another is to vary the shape and location of the slots and their associated couplers, as in a traveling wave antenna. This results in phase and amplitude variations on the slot. An example is shown in figure 6, where the slot spacings have been set at λ, λ, and 0.88λ respectively, and the resulting elevation pattern (figure 7) has a large lobe slightly downtilted and minimized radiation directly downward.

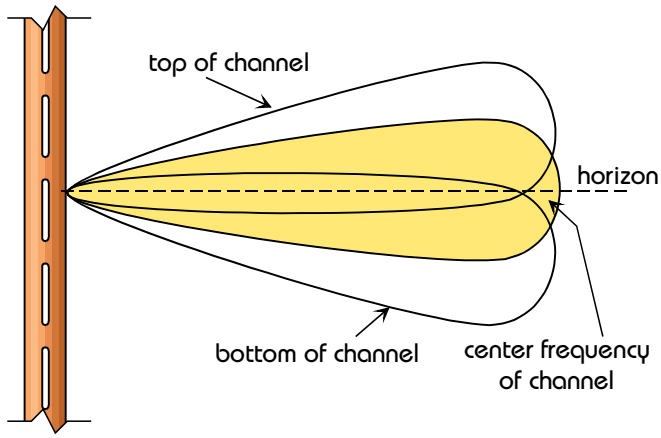


Figure 5. Beam Steering

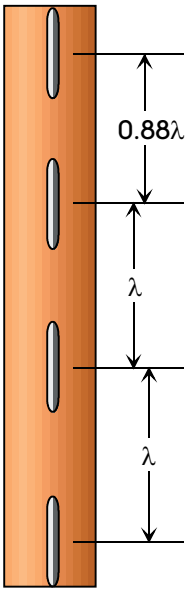


Figure 6. Unequal Slot Spacing

Still another method of shaping the elevation pattern is to divide the antenna into equal multi-bay standing wave sections (for example, 4 bays per section). By feeding each short section at a controlled phase and amplitude,

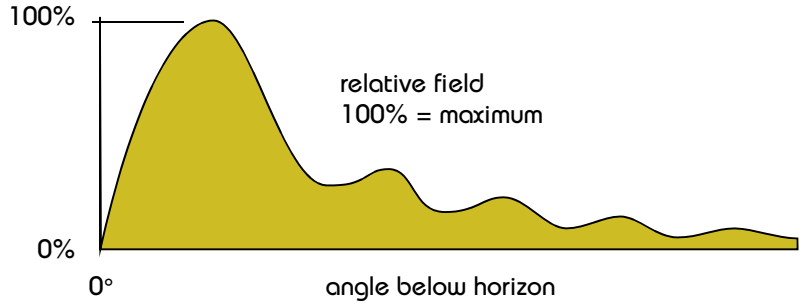


Figure 7. Elevation Pattern Generated by Unequal Slot Spacing

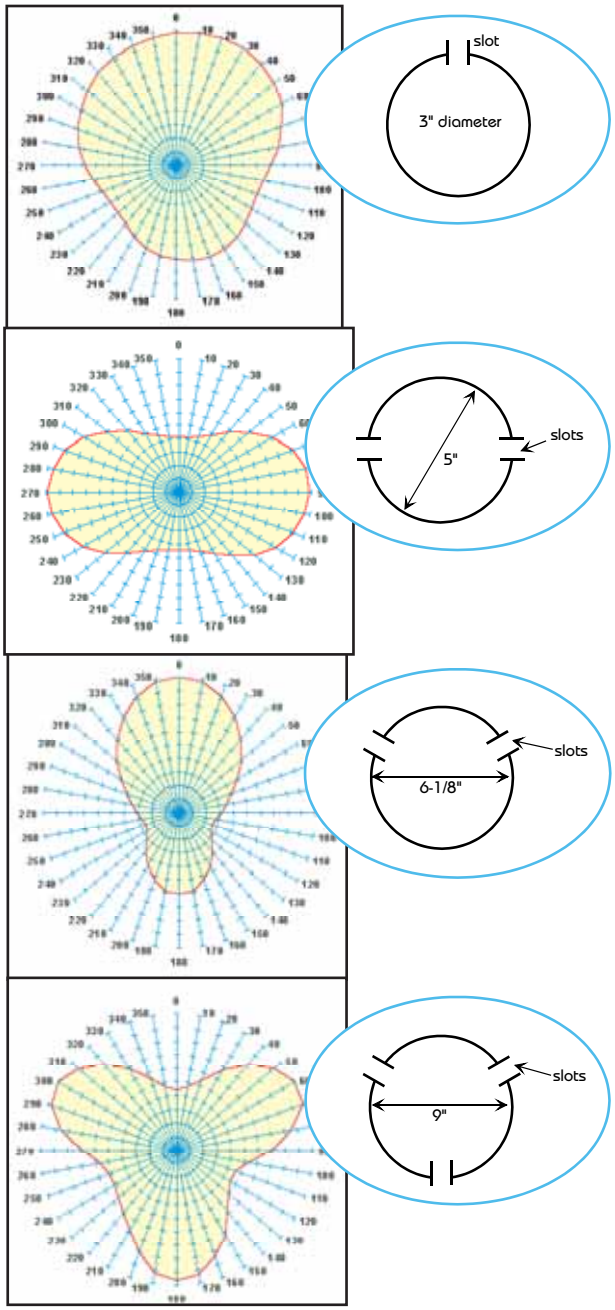


Figure 8. Patterns Formed by Slot Manipulation

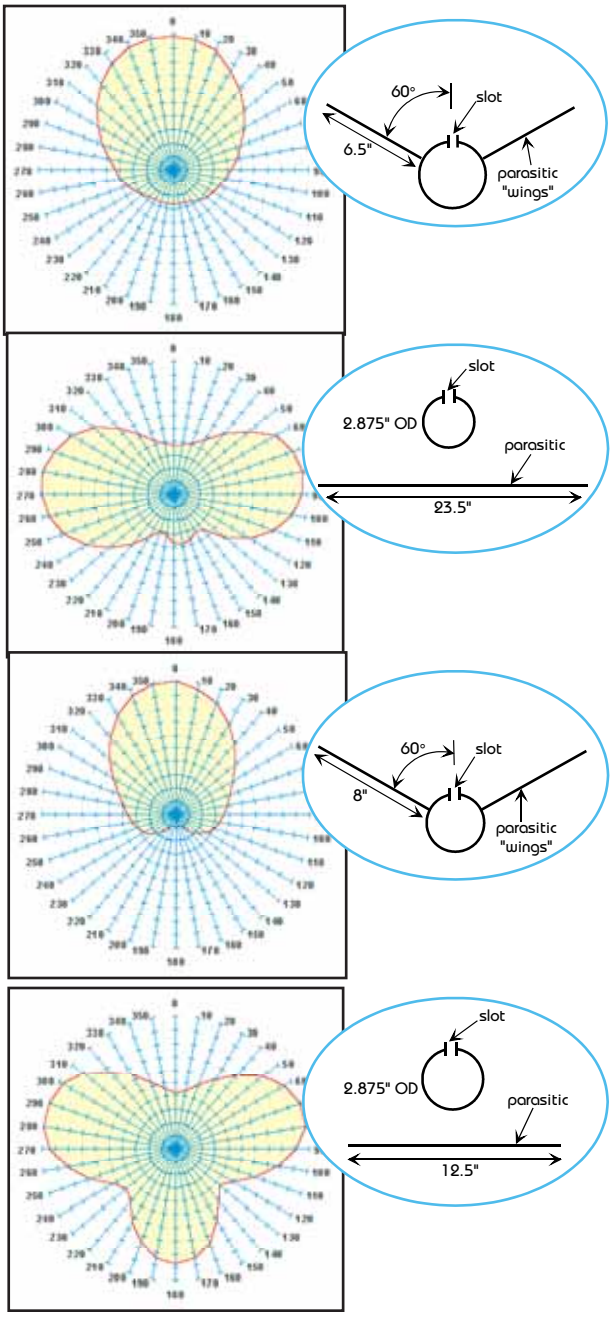


Figure 9. Patterns Formed by the Use of Parasitics

the desired elevation pattern can be achieved. The results are the same as the calculated elevation patterns for large end fed arrays. This method requires a more complex branched feed system. However, because the phase differential across a 6 MHz band is not very large over a length of only four slots, each section can be treated as if it were entirely in-phase, and the resulting array is easy to tune.

Shaping the Azimuth Pattern

The azimuth pattern radiated from a slot antenna can be manipulated in two ways. The first method is to change the configuration of the outer conductor. This includes changing its diameter and/or varying the number and location of its slots. The resulting patterns can be predicted using the Hankel function equations. Figure 8 shows patterns created by simple slot manipulation.

The second method of shaping the azimuth pattern is to attach parasitics at each slot on the outside of the tube. By changing the shape, size and location of the parasitics, different patterns can be generated. The patterns in figure 9 are measured empirically, since with the added complexity of parasitics, the Hankel functions alone are no longer adequate to predict the pattern. Work is being done on ways to predict these patterns mathematically.

Tower Effects

The metal of the supporting tower - indeed, any metal in the aperture of the antenna - affects the broadcast pattern. Therefore, side mount antennas generally cannot provide a true omnidirectional pattern. Figure 10 shows the effect of a tower on an omnioid free space pattern.

There are various ways to minimize this effect:

- Move the antenna far enough away from the tower to minimize the tower's effect. This is usually not practical, for reasons of weight and rigidity.
- Design special tower sections that have no horizontal members in the plane of the antenna and no large diameter vertical members. These tower sections need to be specially designed for each channel and length of antenna and are therefore expensive and sometimes impractical.
- Top-mount the antenna. Pattern distortion due to tower effects does not occur when a slot antenna is top mounted. However, the weight of the antenna increases considerably due to the wall thickness of the outer conductor required to structurally support the antenna.

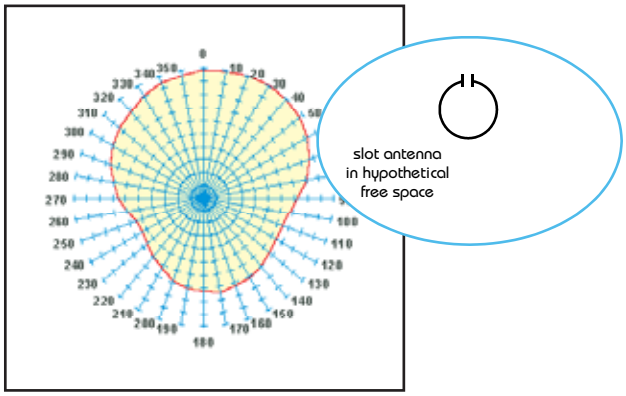


Figure 10A. Measured Free-Space Omnioid Pattern

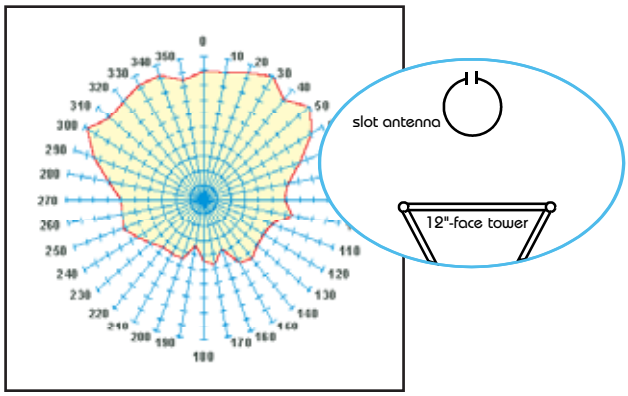


Figure 10B. Measured Omnioid Pattern on 12"-Face Tower

Tower effects on the pattern are measurable. Software programs exist that can be used to predict the tower effects. However, these programs only allow for gross approximations and can not account for all characteristics of the tower, such as ladders and internal coax runs. To obtain accurate, dependable results, measurements must be made by using full size or scale model antennas and towers.

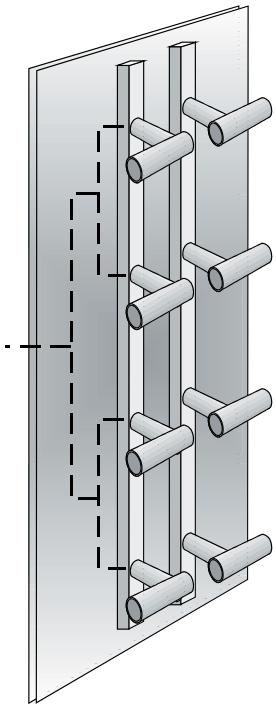


Figure 11. UHF Broadband Dipole Panel with Integrated Parallel Feed System (broken line)

Panel Systems

Broadband UHF panel systems have been used in Europe for more than 30 years. With the advent of DTV and the shortage of tower space in the United States, broadband panel antennas are now becoming more attractive to US broadcasters.

Operating Principle

When a horizontally polarized dipole is placed 1/4-wavelength away from a reflective screen, the backlobe is reduced and the energy is directed forward, creating a higher gain and a narrower azimuth beamwidth. An array of these dipoles over a screen (figure 11) is a "panel." In a panel, half-wave vertical spacing is used to reduce unwanted vertical radiation. A feed system, providing equal amplitude and phase to each dipole, is integrated into the mechanical structure of the panel. Finally, a radome is used to protect the internal structure from the elements.

The shape of the dipoles, the half-wave spacing, and the use of a branched feed system produce a broadband panel with a VSWR of less than 1.1:1 over the entire UHF band.

The azimuth pattern of a single panel will narrow with increased frequency. This is due to the change in wavelength versus the size of the dipole and the spacing off the screen. By using the radome's shape and material to compensate for this pattern narrowing, a constant azimuth beamwidth over the entire band can be achieved.

A panel system consists of a number of low power broadband panels mounted around a tower with an appropriate feed system to create the desired pattern. Patterns change with tower size and the number of panels mounted around the tower.

Advantages

Due to the half-wave spacing and the broadband feed system, a panel system can broadcast the entire UHF band at below a 1.1:1 VSWR.

Due to panel size versus frequency, a panel system can be sidemounted and still achieve an omni-like pattern, although with scalloping (see figure 12).

Panel antennas can also be used to achieve directional patterns. These are achieved by varying the number of panels around the tower, or by varying the power division and phase to each panel (see figures 13 and 14).

Elevation patterns can also be tailored by varying the amount of power and phase to each level, as discussed earlier in the context of slot antennas.

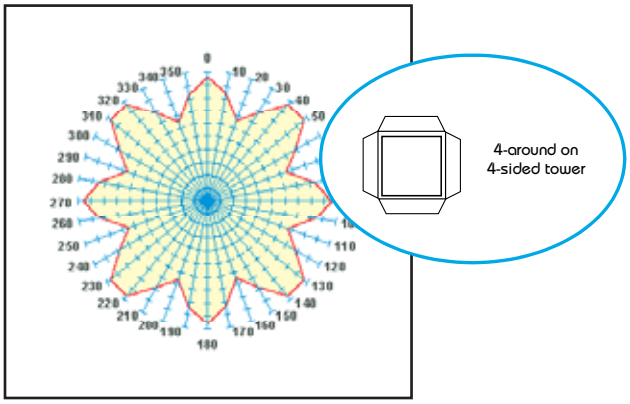


Figure 12. UHF Panel Omni Pattern

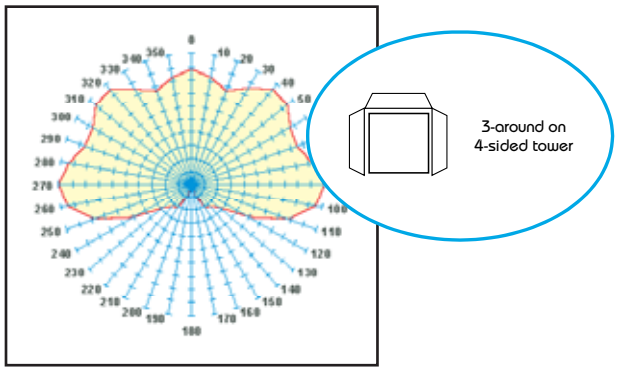


Figure 13. 3-Around Panel on Square Tower

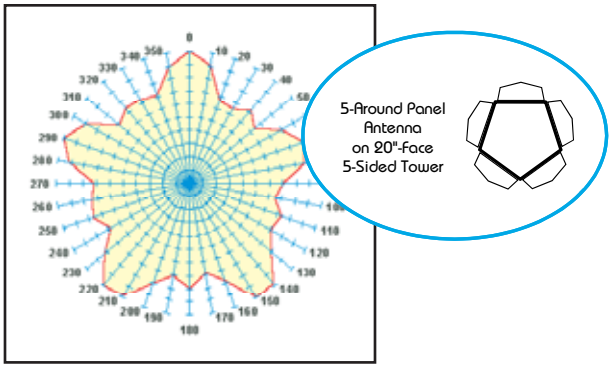


Figure 14. 5-Around Panel on 5-Sided Tower

Concerns

While a single panel can be very broadband, panel system VSWRs can reach undesirable levels. This is caused by the accumulation of component mismatches. VSWR can be reduced by offsetting the panels by 1/4λ (90°) and feeding them out of phase by 90° (fig 15). The 90° phase shift creates a reflection out of phase with the original signal, which results in a canceling of the reflected energy in the feed system and thus produces a low VSWR. The 1/4-wave offset and 90° phase shift result in the signals in the far field adding in-phase.

A VSWR bump can also be produced at the frequency of operation by the feed cable running up the tower. These bumps are caused by equal spacing of transmission line sections, and the addition of the slight mismatches at the mating sections. This can be prevented by making rigid line pieces different lengths, or by running a continuous piece of flex line up the tower.

VSWR is not the only concern with broadband panel systems. In an omnidirectional panel system, as the frequency of operation increases, the scalloping of the azimuth pattern increases. This scalloping can be compensated for over small bandwidths by manipulatiing the configuration and phasing of the individual panels.

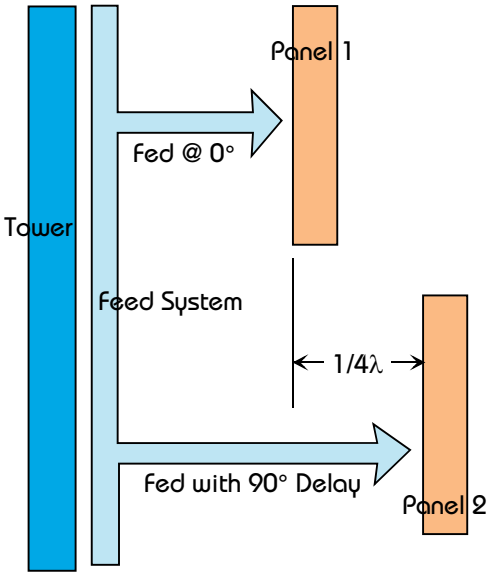


Figure 15. 1/4-Wave Offset Used in Conjunction with 90° Phase Shift

Narrowband Panels

Narrowband panel systems (figure 16) can be used for special low-power television broadcast applications. These are generally used where slot antennas are not appropriate, such as on large towers, or where a complex pattern is required.

Narrowband panels are similar to broadband panels, except they contain a series feed system instead of a parallel one. They can be made physically smaller because they are sized for a specific channel. Their radomes do not have to compensate for pattern narrowing.

Mechanical Structure

Panel systems are large and not self-supporting. They require an external spine to support them and their often complex and bulky feed systems. Panel arrays also produce higher windload than slot arrays

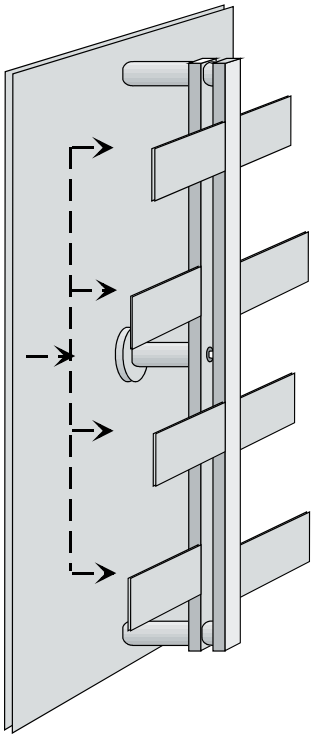


Figure 16. UHF Narrowband Dipole Panel with Integrated Series Feed System (broken line)

of the same gain and pattern shape. Also, internal maintenance access must be designed in for the panels and feed system.

Superturnstile Antennas

The superturnstile antenna has been used to broadcast VHF frequencies for many years. Its use at UHF frequencies, however, is relatively new.

The superturnstile combines the omnidirectionality of a top-mounted slot antenna with the full UHF bandwidth of a panel antenna. Like slot antennas, superturnstiles can be stacked to allow more stations on the same antenna system. This combination of features makes the superturnstile the best choice when both bandwidth and pattern omnidirectionality are required.

Operating Principle

To understand how the superturnstile antenna works, it is first necessary to understand a standard turnstile. In a turnstile, two dipoles, in the horizontal plane and perpendicular to one another, are fed with a 90 progressive phase shift and equal amplitude. The result is that the dipole arms have phases of 0°, 90°, 180°, and 270°. A single dipole will yield a figure-8-shaped azimuth pattern (figure 17). When two dipoles are laid perpendicular to each other (figure 18), their combined azimuth pattern is roughly omnidirectional (figure 19).

Since the dipoles are at right angles and transmitting in-phase, the turnstile is circularly polarized in the vertical direction, and the major lobes of the elevation pattern are directly upward and downward (figure 20). Therefore, we half-wave-space the turnstile radiators vertically, to redirect excessive downward radiation as desirable gain (figure 21).

In a superturnstile, the dipoles are replaced with their complementary slotted sheets, oriented vertically and at right angles to each other, and again fed 90° out of phase and at equal power levels. The slotted sheets are typically 0.7 lambda by 0.5 lambda dimensionally (figure 22). The azimuth patterns of a single slot and two slots at right angles are similar to figures 17 and 19 above.

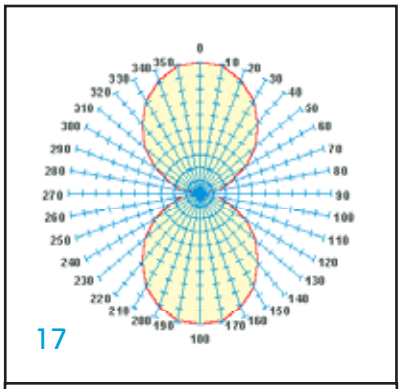
However, unlike a dipole, a slot yields minimal vertical radiation. Thus the elements may be stacked at one wavelength separation, requiring a less complicated feed system.

Advantages

A broadband superturnstile is fed by a parallel network (figure 23). Beam tilt and null fill are achieved by phasing of the stacked elements as with a dipole or slot array.

By adjusting the amplitude and phase of each level, customized elevation patterns can be achieved (figure 24).

Since the superturnstile is generally used where its natural omnidirectional azimuth pattern is desired, it is usually not necessary to manipulate



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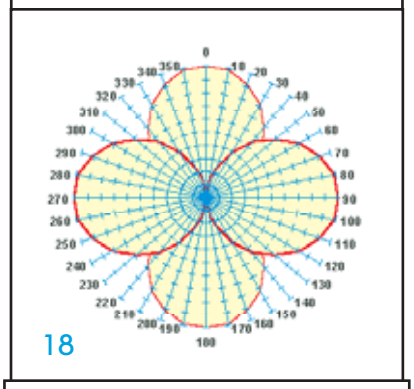

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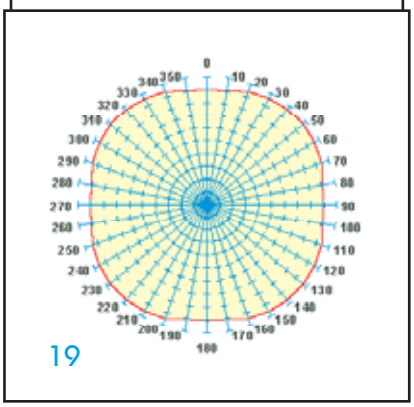

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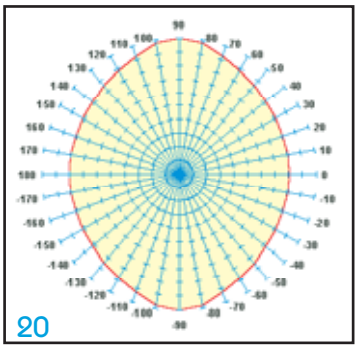

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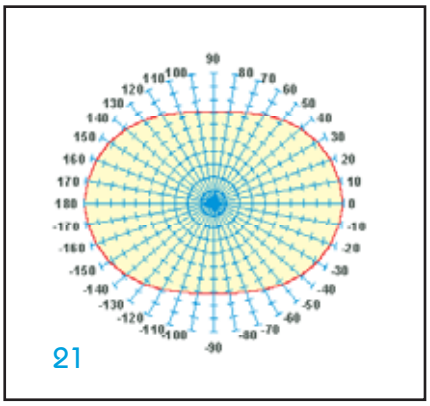

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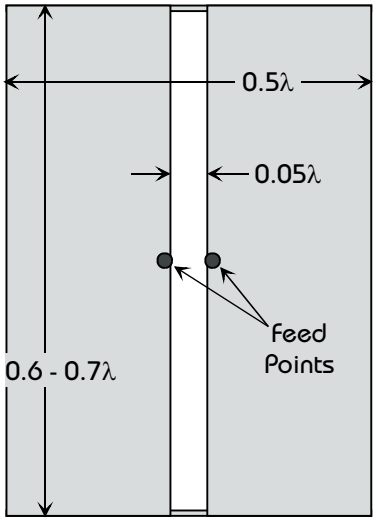

Figure 22. Typical Superturnstile Dimensions

the azimuth pattern (figure 25). Due to the broadband feed system and the hybrids, the superturnstile can broadcast the entire UHF band at or below 1.1:1 VSWR.

Concerns

As with broadband panel systems, the coaxial feedline can cause VSWR problems. Care must be taken to use the correct lengths of rigid coax. Another option is to run a single length of flexible coax.

Mechanical Structure

Since the antenna is made of small radiating elements, it is not self-supporting. Therefore, a superturnstile is enclosed in a radome that supports the antenna as well as providing environmental protection. The parallel feed network is also enclosed in the radome. Access is provided for inspection and servicing, often through hatches at the bottom and top of the radome.

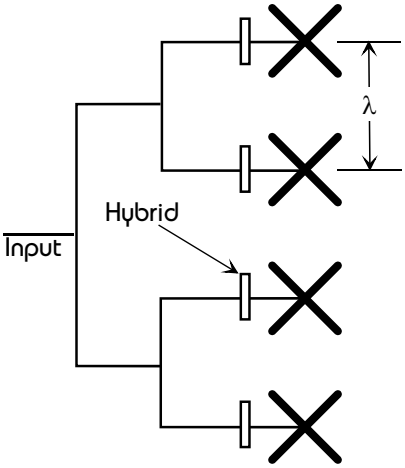

Figure 23. Superturnstile Parallel Feed Network

The weight of a high power superturnstile is about one third of an equivalent top-mounted slot antenna. However, because of the large radome diameter, the superturnstile will generate higher windload.

Conclusion

Figure 26 summarizes the main points of this presentation. This decision tree can be used to help determine which type of antenna is most appropriate for a variety of UHF requirements. Contact the antenna manufacturer to discuss the specific installation before making a final decision.

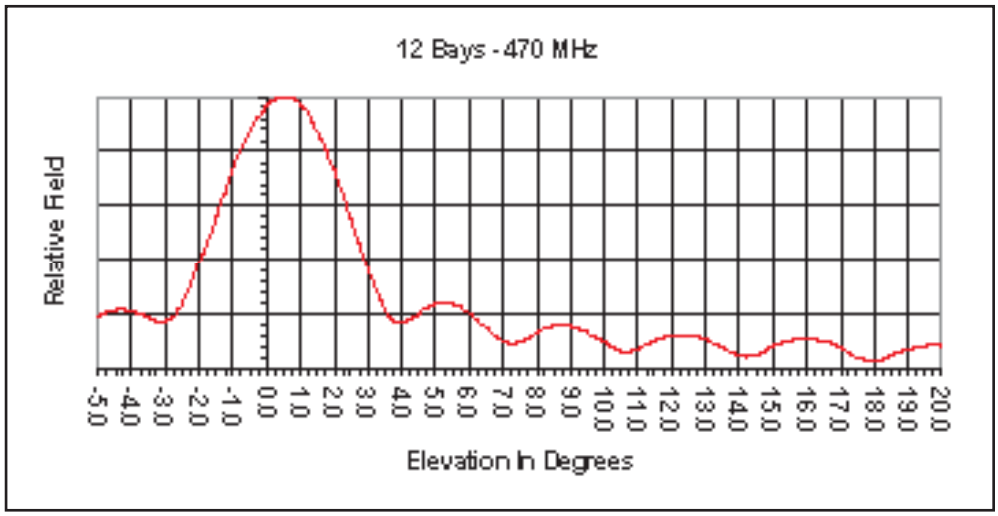

Figure 24. Customized Elevation Pattern of a Superturnstile Antenna

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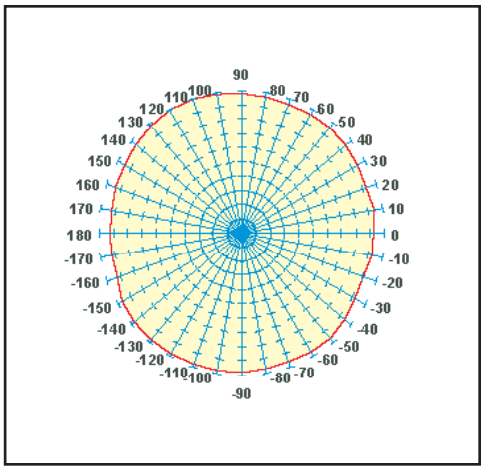

Figure 25. Measured Superturnstile Azimuth Pattern

The Contributors

Dean Casciola is the Senior Design Engineer for Shively Labs of Bridgton, Maine. He has been involved in the design and development of antennas from 54 MHz up to 36 GHz since 1983. He was graduated from Resselaer Polytechnic Institute in 1981, with the degree of Bachelor of Science in Electrical Engineering, and earned

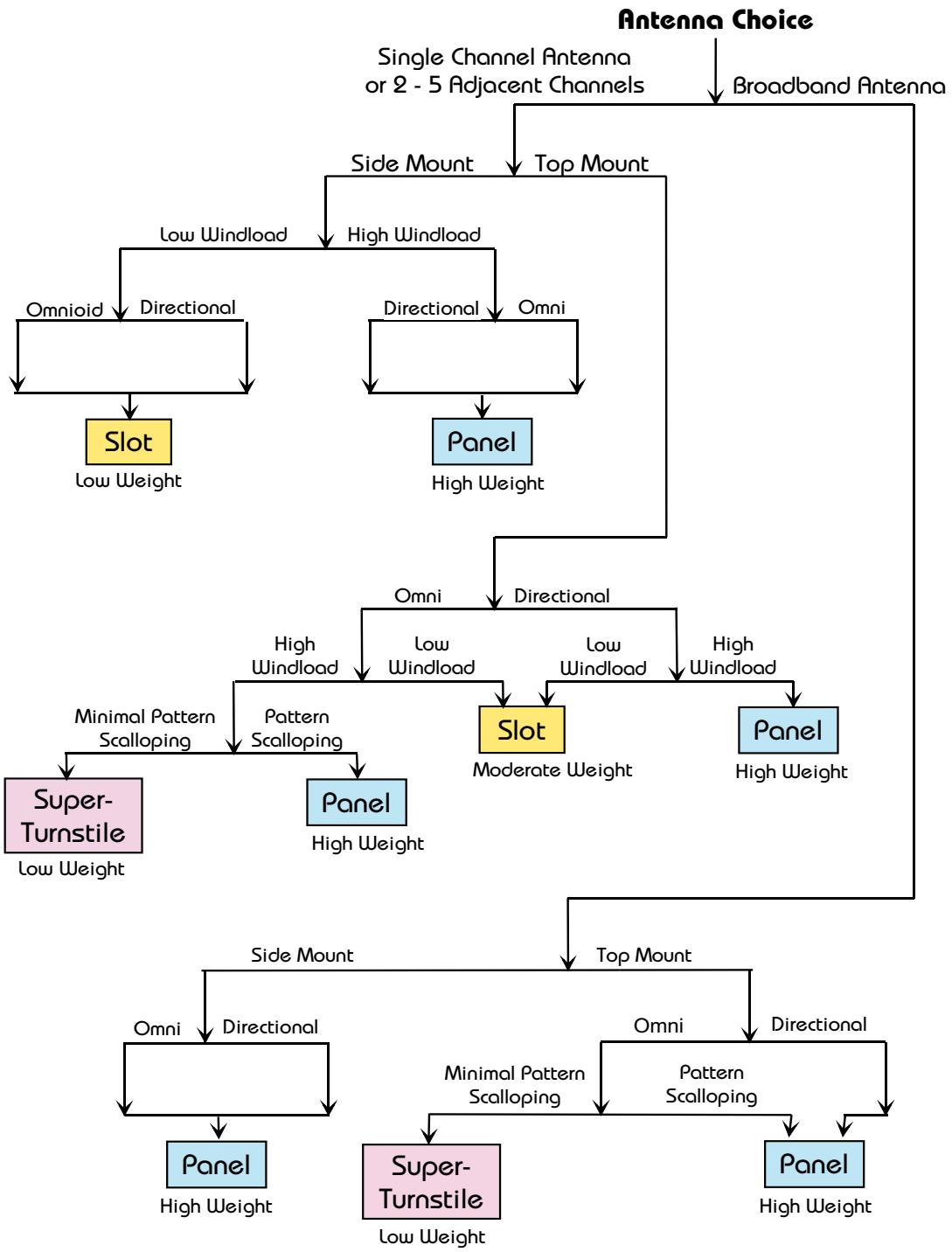

Figure 26. Decision Tree for UHF Antenna Selection

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